Centered Natural-Parameter Group Selection for Posterior-Score Contrast Support in von Mises–Fisher Mixtures

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Abstract

Finite mixtures of von Mises–Fisher distributions are widely used for directional clustering, but existing sparse formulations typically target component prototypes rather than coordinates that separate posterior scores. We define posterior-score contrast support through pairwise posterior log-score contrasts and express it using centered natural parameters. This leads to the Eta-centered group lasso (E-CGL), which applies a coordinate-wise group penalty to centered contrasts while leaving common natural-parameter effects unpenalized; E-ACGL is considered as an adaptive extension. Estimation combines a guarded proximal-gradient generalized-EM path with target-preserving support-constrained refitting and information-criterion selection conditional on the number of components. We establish the centered decomposition, label-invariant support, the closed-form proximal map, and monotonicity of accepted updates. Oracle-error-calibrated simulations show accurate sparse-support recovery at larger sample sizes and identify finite-sample limitations under dense or weak support. In a full-data analysis of the cleaned Classic3 corpus (n = 3,883, d = 2,000), E-CGL selected 1,421 coordinates and attained ARI 0.9761 and NMI 0.9539; the dense free-κ_k fit using all coordinates attained ARI 0.9760 and NMI 0.9538. Supplementary five-split analyses gave a mean conditional support Jaccard index of 0.933 in Classic3, whereas BBCSport showed held-out density loss under sparsification. The method is intended for sparse or compressible posterior-score contrasts rather than as a general replacement for dense or prototype-oriented mixtures.

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1. Introduction

Directional representations arise whenever only the orientation of an observation is meaningful. Common examples include normalized document vectors, compositional profiles after normalization, and embedding vectors on the unit hypersphere. The von Mises-Fisher (vMF) distribution is a standard model for such data, and finite vMF mixtures provide a model-based alternative to spherical k-means (Banerjee et al., 2005). In high dimensions, however, the fitted component structure can involve many coordinates, making both estimation and interpretation difficult.

Sparse vMF mixtures have primarily been formulated through sparsity of the component direction vectors. Rossi and Barbaro (2022), for example, penalize the entries of μ_k to estimate sparse directional prototypes. This target is appropriate when the scientific question concerns which coordinates describe each component prototype. It is not identical to asking which coordinates distinguish components in posterior classification. A coordinate can be active in every prototype and still contribute almost equally to all component scores. Conversely, a coordinate with a modest directional loading can be influential when component concentrations differ. The distinction is therefore between prototype support and the feature-dependent linear support of posterior component comparisons.

This paper targets the latter support through the natural parameters $\eta_k = \kappa_k \mu_k$. For each observed coordinate, the component-specific natural parameters are decomposed into a common baseline and a centered contrast. The proposed E-CGL penalty groups the centered contrast across components, so its unit of regularization is the same coordinate-level unit used to define the target support. The common baseline remains unpenalized. E-ACGL is considered as an adaptive extension with weights fixed from a dense initial fit.

The construction draws on contrast penalties and grouped variable selection, but applies them to a different estimand. Penalized Gaussian mixtures have used entry-wise mean shrinkage, grouped component means, and pairwise mean fusion (Pan and Shen, 2007; Xie et al., 2008; Guo et al., 2010). Sum-to-zero factor contrasts have also been used to separate common levels from heterogeneity (Bondell and Reich, 2009), while other directional-mixture methods target feature saliency, block structure, or hierarchical parameter regularization (Amayri and Bouguila, 2013; Salah et al., 2016; Gopal and Yang, 2014; Taghia et al., 2014; Salah and Nadif, 2019). Discriminative subspace approaches provide another route to variable relevance in model-based clustering (Bouveyron and Brunet, 2012; Bouveyron and Brunet-Saumard, 2014). The contribution here is the alignment of a centered natural-parameter estimand, a coordinate-group penalty, and posterior linear-score interpretation in a vMF mixture.

Candidate supports are generated by a guarded path with proximal gradient updates inside generalized EM. They are compared after support-constrained refitting that preserves the target. The primary selector is a practical BIC computed from the refitted likelihood; alternative degrees of freedom and EBIC rules are treated as sensitivity analyses. Support-recovery claims condition on fixed K. When K is unknown, component-number selection and support selection are examined in separate stages rather than through a single joint criterion.

The empirical study has three parts. Oracle-error calibrated simulations compare posterior-score support recovery across sample size, ambient dimension, and concentration heterogeneity. Shared-background and dense-support designs separate the proposed estimand from prototype sparsity and record settings in which sparse posterior-score support is inappropriate. Locked repeated-holdout analyses of Classic3 and BBCSport examine clustering, coordinate reduction, and token-level contrasts on vocabulary-aligned directional representations.

The contributions are therefore: (i) a posterior linear-score support estimand for vMF mixtures and its centered representation; (ii) E-CGL and its adaptive extension, together with a guarded estimation and exact-refit procedure; and (iii) an empirical evaluation that distinguishes the intended sparse-support regime from prototype-oriented and dense-support settings. E-CGL is not intended to replace sparse-prototype methods when prototype sparsity is the scientific target.

The remainder of the paper is organized as follows. Section 2 presents the vMF model, posterior-score contrast support, and centered group regularization. Section 3 describes the estimation algorithm, pathwise selection and refitting, and the structural and algorithmic properties used in the analysis. Sections 4 and 5 report the numerical studies and locked real-data analyses, respectively. Section 6 concludes with the empirical scope and limitations of the method.

2. Model and centered regularization

2.1. vMF mixture in natural parameters

Let $x_i \in \mathbb{S}^{d-1}$, $i = 1, \ldots, n$, denote unit-normalized directional observations. A K-component vMF mixture has density

$$f(x_i; \Theta) = \sum_{k=1}^{K} \pi_k C_d(\kappa_k) \exp\{\kappa_k \mu_k^\top x_i\},$$

where $\pi_k > 0$, $\sum_k \pi_k = 1$, $\|\mu_k\|_2 = 1$, and $\kappa_k \geq 0$. Its normalizing constant is

$$C_d(\kappa) = \frac{\kappa^{d/2-1}}{(2\pi)^{d/2} I_{d/2-1}(\kappa)}.$$

At $\kappa = 0$, this expression is interpreted by continuous extension as $C_d(0) = \Gamma(d/2)/(2\pi^{d/2})$.

Writing $\eta_k = \kappa_k \mu_k \in \mathbb{R}^d$ gives

$$f(x_i; \Theta) = \sum_{k=1}^{K} \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i).$$

The component-level mapping back to (μ_k, κ_k) and its zero- concentration exception are recorded in Section 3.4. The observed log-likelihood is

$$\ell(\Theta) = \sum_{i=1}^{n} \log \left\{ \sum_{k=1}^{K} \pi_k C_d(\|\eta_k\|_2) \exp(\eta_k^\top x_i) \right\},$$

2.2. Posterior-score contrast support

The component score

$$s_k(x) = \log \pi_k + \log C_d(\|\eta_k\|_2) + \eta_k^\top x$$

implies the pairwise posterior log-odds representation

$$\log \frac{\Pr(Z = k \mid x)}{\Pr(Z = \ell \mid x)} = a_{k\ell} + (\eta_k - \eta_\ell)^\top x,$$

$$a_{k\ell} = \log \frac{\pi_k}{\pi_\ell} + \log \frac{C_d(\|\eta_k\|_2)}{C_d(\|\eta_\ell\|_2)}.$$

Thus $\eta_k - \eta_\ell$, rather than $\mu_k - \mu_\ell$, is the coefficient of the feature-dependent linear term. For coordinate j, let

$$H_K = I_K - K^{-1}\mathbf{1}_K\mathbf{1}_K^\top, \qquad c_{\cdot j} = H_K \eta_{\cdot j} = \eta_{\cdot j} - \bar{\eta}_j \mathbf{1}_K,$$

where $\bar{\eta}_j = K^{-1}\sum_k \eta_{kj}$. We define

$$S_\eta = \{j : \|c_{\cdot j}\|_2 > 0\} = \bigcup_{k<\ell} \{j : \eta_{kj} - \eta_{\ell j} \neq 0\}.$$

We use *posterior-score contrast support* for this feature-dependent linear support. It does not include the intercepts $a_{k\ell}$. A coordinate with $c_{\cdot j} = 0$ cancels from every pairwise linear term, although its common baseline $\bar{\eta}_j$ remains in the fitted component densities and may affect $a_{k\ell}$ through the concentration norms. Proposition 1 gives the unique centered decomposition and its pairwise-dispersion identity.

Accordingly, $c_{\cdot j} \neq 0$ defines a decision coordinate; $c_{\cdot j} = 0$ with $\bar{\eta}_j \neq 0$ defines a common coordinate; and $c_{\cdot j} = 0$ with $\bar{\eta}_j = 0$ defines a globally null coordinate. These categories distinguish the proposed target from prototype support, which records nonzero component coordinates without requiring a difference between components.

The support is invariant to permutations of component labels because such a permutation only reorders the entries of $c_{\cdot j}$. It is not invariant to an arbitrary orthogonal rotation of the feature axes. As with other coordinate-selection procedures, its interpretation is therefore relative to the chosen coordinate system.

2.3. Centered group regularization (E-CGL)

For fixed K, the primary penalty is

$$P_{\mathrm{CGL}}(\eta) = \lambda_\eta \sum_{j=1}^{d} \|H_K \eta_{\cdot j}\|_2 = \lambda_\eta \sum_{j=1}^{d} \|c_{\cdot j}\|_2.$$

Each group is one observed coordinate across the K component contrasts, so the unit of regularization matches the definition of S_η. The common baseline lies in the null space of H_K and is not penalized. The first expression writes the penalty as a function of η, whereas the second displays its centered-contrast target. This coordinate-group construction follows the grouped-variable regularization principle of Yuan and Lin (2006). By contrast,

$$\|\eta_{\cdot j}\|_2^2 = K\bar{\eta}_j^2 + \|c_{\cdot j}\|_2^2,$$

so a raw-η group penalty combines common and contrast effects. A direction-space penalty targets prototype coordinates and, when the κ_k differ, does not directly target the coefficients $\eta_k - \eta_\ell$ in the posterior log-odds. These are distinct estimands, not competing parameterizations of the same support.

An entry-wise centered penalty selects component-coordinate entries separately: $\lambda_\eta \sum_{k,j} |c_{kj}|$. E-CGL instead selects the complete coordinate contrast as one group. The resulting proximal map and label invariance are stated in Propositions 3 and 2.

2.4. Adaptive extension

E-ACGL uses

$$P_{\mathrm{ACGL}}(\eta) = \lambda_\eta \sum_{j=1}^{d} w_j \|H_K \eta_{\cdot j}\|_2 = \lambda_\eta \sum_{j=1}^{d} w_j \|c_{\cdot j}\|_2,$$

with weights fixed from the dense initial fit:

$$w_j^{\mathrm{raw}} = \left(\|c_{\cdot j}^{\mathrm{init}}\|_2 + \epsilon\right)^{-\gamma}, \qquad w_j = \frac{w_j^{\mathrm{raw}}}{\mathrm{median}_{1\le h\le d}\, w_h^{\mathrm{raw}}}.$$

The numerical analyses use $\gamma = 1$ and $\epsilon = 10^{-6}$, and the weights remain fixed throughout optimization and path selection. E-CGL, with $w_j = 1$, is the primary estimator; E-ACGL is an adaptive extension and no oracle-property claim is made. This weighting follows the adaptive regularization principle of Zou (2006), applied here to centered coordinate groups.

3. Estimation and properties

3.1. Penalized objective and component updates

For fixed K and λ_η, the path-generating criterion is

$$
\begin{aligned}
\mathcal{L}_{\lambda_\eta}(\Theta) &= \ell(\Theta) - \lambda_\eta \sum_{j=1}^{d} w_j \|H_K \eta_{\cdot j}\|_2 \\
&= \ell(\Theta) - \lambda_\eta \sum_{j=1}^{d} w_j \|c_{\cdot j}\|_2.
\end{aligned} \tag{1}
$$

where $w_j = 1$ for E-CGL and the E-ACGL weights are fixed from the dense initial fit. The penalty and the vMF normalizing constant are coupled through η_k, so the natural-parameter update is computed by a guarded proximal-gradient generalized M-step.

3.1.1. E-step and fixed-responsibility subproblem

At outer iteration t, compute $\tau_{ik}^{(t)}$ and

$$
\tau_{ik}^{(t)} = \frac{\pi_k^{(t)} C_d(\|\eta_k^{(t)}\|_2) \exp\{(\eta_k^{(t)})^\top x_i\}}{\sum_{h=1}^{K} \pi_h^{(t)} C_d(\|\eta_h^{(t)}\|_2) \exp\{(\eta_h^{(t)})^\top x_i\}}, \tag{2}
$$

and define

$$
N_k^{(t)} = \sum_{i=1}^{n} \tau_{ik}^{(t)}, \qquad r_k^{(t)} = \sum_{i=1}^{n} \tau_{ik}^{(t)} x_i.
$$

Conditional on these responsibilities, the natural-parameter block minimizes

$$
F_t(\eta) = f_t(\eta) + g(\eta),
$$

$$
\begin{aligned}
f_t(\eta) &= -\sum_{k=1}^{K} \left\{ (r_k^{(t)})^\top \eta_k + N_k^{(t)} \log C_d(\|\eta_k\|_2) \right\}, \\
g(\eta) &= \lambda_\eta \sum_{j=1}^{d} w_j \|H_K \eta_{\cdot j}\|_2.
\end{aligned}
$$

The smooth term is convex because $-\log C_d(\|\eta_k\|_2)$ is the vMF log-partition function. Its gradient is

$$
\nabla_{\eta_k} f_t(\eta) = N_k^{(t)} A_d(\|\eta_k\|_2) \frac{\eta_k}{\|\eta_k\|_2} - r_k^{(t)},
$$

with the continuous value $-r_k^{(t)}$ at $\eta_k = 0$, where A_d is the vMF mean-resultant-length function.

3.1.2. Centered proximal-gradient update

Let $J_K = K^{-1}\mathbf{1}_K\mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For a step size $s > 0$, set

$$v = \eta^{(m)} - s\nabla f_t(\eta^{(m)}).$$

The update separates by observed coordinate and has the closed form

$$\eta_{\cdot j}^{(m+1)} = J_K v_{\cdot j} + \left(1 - \frac{s\lambda_\eta w_j}{\|H_K v_{\cdot j}\|_2}\right)_+ H_K v_{\cdot j}. \tag{3}$$

The common component is therefore retained, while the centered contrast is group-thresholded; this is a proximal group soft-thresholding operation (Parikh and Boyd, 2014). Mixing proportions are updated by

$$\pi_k^{\text{cand}} = \frac{\max(N_k^{(t)}/n, 10^{-12})}{\sum_h \max(N_h^{(t)}/n, 10^{-12})}.$$

The initial step size is $s = 1/\max_k N_k^{(t)}$. If the smooth majorization condition

$$f_t(\eta^{(m+1)}) \le f_t(\eta^{(m)}) + \langle \nabla f_t(\eta^{(m)}), \eta^{(m+1)} - \eta^{(m)}\rangle + \frac{\|\eta^{(m+1)} - \eta^{(m)}\|_F^2}{2s} \tag{4}$$

is not satisfied within numerical tolerance, s is halved and the proximal proposal is recomputed. This backtracking is repeated for at most 40 trials.

3.2. Guarded proximal generalized-EM algorithm

After the inner proximal-gradient update, the implementation evaluates the penalized auxiliary function and the observed penalized log-likelihood. The candidate is retained only if neither quantity decreases by more than 10^{-8}. Otherwise, the path fit stops, returns the previous accepted estimate, and records a failure diagnostic. Convergence is declared when the relative change in the observed penalized criterion is below 10^{-7}.

Components with effective count below 10^{-8} are retained to preserve the fixed value of K and the event is recorded. They are not pruned or reinitialized. These safeguards support statements about accepted numerical updates; they do not establish convergence of the complete parameter sequence, stationarity of every fitted path point, or global optimality.

Algorithm 1 summarizes the complete pathwise procedure. The same algorithm fits E-CGL and E-ACGL; only the fixed weights differ.

Algorithm 1. Guarded path algorithm for E-CGL and E-ACGL

Input: Data X, component number K, path size L, method indicator, iteration limits, and tolerances $\varepsilon_{\text{conv}}, \varepsilon_{\text{acc}}$.

Output: Selected support $\widehat{S}$, its generating tuning set $\widehat{\Lambda}_\eta$, representative $\widehat{\lambda}_\eta = \max \widehat{\Lambda}_\eta$; refitted parameters $\widehat{\Theta}^{\text{refit}}$; numerical diagnostics.

Stage 1: Dense start and path construction

Dense start: fit the finite-κ dense model from multiple initializations and retain the converged fit with the largest observed log-likelihood.

Weights: set $w_j = 1$ for E-CGL; for E-ACGL, compute w_j from $\Theta^{(0)}$ and then hold w fixed.

Path: construct the KKT-geometric sequence $\Lambda = (0, \lambda_1, \ldots, \lambda_{L-1})$ from dense to sparse.

Stage 2: Guarded penalized path

for $\lambda_\eta \in \Lambda$ **do**; warm-start from the preceding accepted path fit.

  repeat

    E-step: evaluate τ_{ik} by (2); set $N_k \leftarrow \sum_i \tau_{ik}$, $r_k \leftarrow \sum_i \tau_{ik} x_i$.

    M-step: set $\pi_k^+ \leftarrow N_k/n$ and η^+ by (3).

    Step control: recompute the candidate while halving s until (4) and both acceptance inequalities below hold, or the backtracking limit is reached.

    Acceptance: retain Θ^+ if $\Delta Q_{\lambda_\eta} \geq -\varepsilon_{\text{acc}}$ and $\Delta \mathcal{L}_{\lambda_\eta} \geq -\varepsilon_{\text{acc}}$; after the backtracking limit, otherwise mark the path point as failed and terminate the path.

  until $|\Delta \mathcal{L}_{\lambda_\eta}|/(1 + |\mathcal{L}_{\lambda_\eta}|) < \varepsilon_{\text{conv}}$ or the iteration limit is reached.

  Store: S_{λ_η}, the fitted criterion, and diagnostics.

end for

Stage 3: Support-constrained refit and selection

Refit: for each distinct S, compute $\widehat{\Theta}_S^{\text{refit}}$ subject to $H_K \eta_{\cdot j} = 0$ for $j \notin S$.

Score: $\text{BIC}^{\text{refit}}(S) \leftarrow -2\ell(\widehat{\Theta}_S^{\text{refit}}) + \log(n)\,\text{df}(S)$.

Select: $\widehat{S} \leftarrow \arg\min_S \text{BIC}^{\text{refit}}(S)$.

Return: $\widehat{S}$, $\widehat{\Lambda}_\eta$, $\widehat{\lambda}_\eta = \max \widehat{\Lambda}_\eta$, $\widehat{\Theta}^{\text{refit}}$, and diagnostics.

3.3. Pathwise selection and support-constrained refit

3.3.1. KKT-geometric path

The path starts at $\lambda_\eta = 0$ and uses warm starts. Responsibilities from the unpenalized dense initializer define (N_k, r_k). Let $r_\bullet = \sum_k r_k$ and $N_\bullet =$

$\sum_k N_k = n$, and let η^{common} be the pooled finite-cap vMF estimate formed from $(N_\bullet, r_\bullet)$. The matrix η_0 repeats η^{common} across components. A KKT-scale upper endpoint for this initial fixed-responsibility subproblem is

$$\lambda_{\max} = 1.05 \max_{1 \le j \le d} \frac{\|H_K \nabla f_t(\eta_0)_{\cdot j}\|_2}{w_j}.$$

The confirmatory path contains zero and 119 positive values on a logarithmic grid from $0.01\lambda_{\max}$ to $\lambda_{\max}$. It stops early if the active support has at most one coordinate. The principal simulations therefore use at most 120 path points.

3.3.2. Target-preserving support-constrained refit

The penalized path identifies candidate supports, but the final estimates are obtained by a sparsity-penalty-free, finite-cap constrained refit. For a candidate support S, impose

$$j \notin S \quad \Longrightarrow \quad c_{\cdot j} = 0 \quad \Longleftrightarrow \quad \eta_{1j} = \cdots = \eta_{Kj} = \bar{\eta}_j.$$

Thus, an inactive posterior-score coordinate retains an estimated common baseline in the natural-parameter space. The constrained parameterization is

$$\eta_{kj} = b_j + c_{kj}, \qquad \sum_{k=1}^{K} c_{kj} = 0, \qquad c_{kj} = 0 \quad (j \notin S).$$

For each active coordinate, the first $K-1$ contrasts are free and the last is set to their negative sum. Conditional on the responsibilities, the M-step maximizes

$$Q_S(\eta) = \sum_{k=1}^{K} \left[\eta_k^\top r_k + N_k \log C_d(\|\eta_k\|_2)\right]$$

over $b \in \mathbb{R}^d$ and the free active contrasts. The implementation uses gradient calculations consistent with the numerically evaluated vMF normalizer and L-BFGS-B, followed by step-halving if the observed log-likelihood decreases. Responsibilities, mixing proportions, and the constrained natural parameters are iterated until the numerical stopping criterion is met.

The target-preserving support-constrained refit enforces the centered support constraint directly while retaining the common baseline. The constrained likelihood is optimized numerically and is not claimed to attain the global maximum of the non-convex mixture likelihood.

The earlier active-only refit imposed $\eta_{\cdot j} = 0$ outside S and therefore removed both the centered contrast and the common baseline. It is retained only as an ablation diagnostic; the target-preserving support-constrained refit is used in the primary analyses.

3.3.3. Support selection by BIC after refit

Let $m_S = |S|$. The practical degrees-of-freedom approximation is

$$\mathrm{df}(S) = d + (K-1)m_S + (K-1)\mathbf{1}(m_S > 0).$$

The three terms count the coordinate-wise common baselines, the active centered contrasts, and the mixing proportions when nonzero component contrasts are present. This is a model-selection approximation, not an exact effective degrees of freedom for a penalized non-regular mixture.

For each unique support on the path, define

$$\mathrm{BIC}^{\mathrm{refit}}(S) = -2\ell(\widehat{\Theta}_S^{\mathrm{refit}}) + \log(n)\,\mathrm{df}(S).$$

The final support is

$$\widehat{S} = \arg\min_{S \in \mathcal{S}_{\mathrm{path}}} \mathrm{BIC}^{\mathrm{refit}}(S).$$

Several tuning values may generate the same support, so BIC after refit does not identify a unique tuning value. We report

$$\widehat{\Lambda}_\eta = \{\lambda_\eta : S_{\lambda_\eta} = \widehat{S}\}, \qquad \widehat{\lambda}_\eta = \max \widehat{\Lambda}_\eta$$

as a deterministic representative convention.

All distinct supports encountered on the path are refitted. The information criterion follows the BIC principle of Schwarz (1978), using the practical degrees-of-freedom approximation above.

An EBIC sensitivity criterion is

$$\mathrm{EBIC}_\gamma(S) = \mathrm{BIC}^{\mathrm{refit}}(S) + 2\gamma \log \binom{d}{m_S}, \qquad \gamma \in \{0.25, 0.5, 1\}.$$

EBIC (Chen and Chen, 2008) and alternative degrees-of-freedom rules are treated as sensitivity analyses rather than changes to the primary selector.

3.3.4. Selection of the number of components

Component-number selection is not part of the E-CGL support estimator. The number of components and the sparsity level are therefore selected in separate stages. In support-recovery simulations, K is fixed at the data-generating value so that support selection is not confounded with component-number misspecification.

For an unknown K, let $\mathcal{K}$ be a prespecified candidate set. The practical two-stage procedure is:

1. Fit an unpenalized dense vMF mixture for each $K \in \mathcal{K}$ using multiple starts.
2. Compare candidate K values using label-free density criteria, information criteria, and partition stability as appropriate to the application.
3. Fix the selected component resolution $\widehat{K}$.
4. Fit the E-CGL path at $\widehat{K}$ and choose a distinct support by BIC after the target-preserving support-constrained refit.

Symbolically,

$$\widehat{K} = \arg\min_{K \in \mathcal{K}} \mathcal{C}_{\text{dense}}(K), \qquad \widehat{S}_{\widehat{K}} = \arg\min_{S \in \mathcal{S}_{\text{path}}(\widehat{K})} \text{BIC}^{\text{refit}}(\widehat{K}, S).$$

Here $\mathcal{C}_{\text{dense}}$ may use independent or held-out negative log-likelihood, bootstrap out-of-bag negative log-likelihood, or an information criterion. No single information criterion is asserted to select the correct resolution in every high-dimensional mixture. When density fit and stability favor different K, both competing resolutions are reported.

3.3.5. Computational implementation

The implementation uses log-sum-exp stabilization in the E-step, warm starts along the penalty path, and explicit convergence and constraint checks in the centered-support refit. Optional Rcpp helpers replace low-level numerical operations without changing the statistical algorithm; R-only and Rcpp-helper outputs were checked to numerical tolerance before runtime comparisons.

The current computational claims are limited to empirical equality and runtime diagnostics. The method does not rely on an Rcpp implementation for its statistical definition.

The likelihood of a finite vMF mixture can be unbounded under component concentration collapse (Ng, 2023). We therefore use the same finite numerical parameter space in dense initialization, the penalized path, and support-constrained refitting:

$$0 \leq \kappa_k = \|\eta_k\|_2 \leq \texttt{kappa\_cap} = 10^6.$$

Backtracking rejects any proposed E-CGL or E-ACGL iterate outside this space. A converged refit that reaches 99% of the bound is retained for diagnosis but is not eligible for information-criterion selection. This guard defines the finite computational problem; it is not an additional sparsity penalty. "Exact refit" refers to exact enforcement of the centered-support constraint, not to a global mixture optimum. The algorithmic results below do not claim convergence to a global maximizer.

3.4. Structural properties

The following statements justify the centered support estimand for a fixed number of components. They do not imply uniqueness of a fitted mixture, which remains subject to finite-mixture identifiability and label permutation. Full proofs are given in Appendix A.

Proposition 1 (Centered decomposition and pairwise support)**.** *For any $v \in \mathbb{R}^K$, there is a unique decomposition*

$$v = \bar{v}\mathbf{1}_K + c, \qquad \bar{v} = K^{-1}\mathbf{1}_K^\top v, \qquad \mathbf{1}_K^\top c = 0.$$

For $v = \eta_{\cdot j}$, the following statements are equivalent:

$$c = 0; \qquad \eta_{1j} = \cdots = \eta_{Kj}; \qquad \eta_{kj} - \eta_{\ell j} = 0 \quad \text{for all } k < \ell.$$

Moreover,

$$\|c\|_2^2 = \frac{1}{K}\sum_{k<\ell}(\eta_{kj} - \eta_{\ell j})^2.$$

Consequently,

$$S_\eta = \bigcup_{k<\ell}\{j : \eta_{kj} - \eta_{\ell j} \neq 0\},$$

the union of the feature-dependent linear supports in all pairwise posterior log-odds.

Proposition 1 clarifies the scope of the estimand. If $c_{\cdot j} = 0$, the coefficient of x_j cancels from every pairwise posterior log-odds. The common baseline is nevertheless retained in η_k and may affect the intercept through $\|\eta_k\|_2$; it is therefore not irrelevant to the component density.

Proposition 2 (Label equivariance and support invariance)**.** *Let P be a $K \times K$ permutation matrix. Then*

$$H_K P = P H_K, \qquad \|H_K P \eta_{\cdot j}\|_2 = \|H_K \eta_{\cdot j}\|_2.$$

The E-CGL penalty is invariant to component relabeling, its proximal map is equivariant, and S_η is unchanged.

For a nonzero natural parameter, the component representation is recovered as $\kappa_k = \|\eta_k\|_2$ and $\mu_k = \eta_k / \|\eta_k\|_2$. At $\eta_k = 0$, the vMF component is uniform and its direction is not identified. These algebraic statements do not establish finite-mixture identifiability or uniqueness of the fitted mixture. Component-specific interpretation assumes distinct component distributions and positive mixing weights and is made only up to label permutation. If two components have identical natural parameters, their individual mixing weights are not separately identified. Support-recovery statements are conditional on fixed K.

3.5. Algorithmic properties

Proposition 3 (Closed-form proximal update)**.** *Let $J_K = K^{-1} \mathbf{1}_K \mathbf{1}_K^\top$ and $H_K = I_K - J_K$. For $a \geq 0$ and $v \in \mathbb{R}^K$,*

$$\operatorname{prox}_{a\|H_K \cdot\|_2}(v) = J_K v + \left(1 - \frac{a}{\|H_K v\|_2}\right)_+ H_K v,$$

where the second term is defined as zero when $H_K v = 0$.

The common baseline is unchanged. Group soft thresholding is applied only to the centered contrast, matching the coordinate-level definition of S_η.

Proposition 4 (Smooth majorization and proximal descent)**.** *Fix responsibilities with $N_k > 0$ and fixed weights $w_j > 0$. The function f_t in Section 3.1.1 is convex and has a Lipschitz gradient with constant $L \leq \max_k N_k$. Hence, for $s \leq 1/L$, the proximal-gradient update in Section 3.1.2 does not increase F_t in exact arithmetic. More generally, an update accepted with majorization tolerance $\varepsilon_{\mathrm{maj}}$ satisfies*

$$F_t(\eta^{(m+1)}) \le F_t(\eta^{(m)}) + \varepsilon_{\mathrm{maj}}.$$

Proposition 4 motivates the backtracking rule. The implementation starts from $s = 1/\max_k N_k$ and halves the step until the majorization inequality is satisfied numerically. This check also protects against error from numerical evaluation of the vMF normalizing constant.

Let $Q(\Theta \mid \tau^{(t)})$ denote the expected complete-data log-likelihood under the current responsibilities, and define

$$Q_{\lambda_\eta}(\Theta \mid \tau^{(t)}) = Q(\Theta \mid \tau^{(t)}) - \lambda_\eta \sum_{j=1}^{d} w_j \|H_K \eta_{\cdot j}\|_2.$$

Proposition 5 (Penalized generalized-M monotonicity)**.** *Let $\tau^{(t)}$ be the posterior responsibilities under $\Theta^{(t)}$. If a candidate $\Theta^{(t+1)}$ satisfies*

$$Q_{\lambda_\eta}(\Theta^{(t+1)} \mid \tau^{(t)}) \ge Q_{\lambda_\eta}(\Theta^{(t)} \mid \tau^{(t)}),$$

then, in exact arithmetic,

$$\mathcal{L}_{\lambda_\eta}(\Theta^{(t+1)}) \ge \mathcal{L}_{\lambda_\eta}(\Theta^{(t)}).$$

The implementation verifies both inequalities directly and permits decreases of at most 10^{-8}, so its finite trace is only numerically near-monotone. For the exact-arithmetic algorithm with zero acceptance tolerance, a bounded penalized criterion has a nondecreasing and therefore convergent sequence of objective values. These results do not establish boundedness or convergence of the full parameter sequence, stationary-point convergence, or global optimality.

4. Numerical studies

4.1. Design and evaluation criteria

The primary experiment evaluates recovery of posterior-score contrast support under varying overlap and concentration heterogeneity. Observations are drawn from a balanced $K = 4$ vMF mixture in $d = 200$ dimensions. The coordinates are partitioned as

$$\{1, \dots, d\} = S_C \cup S_D \cup S_N, \qquad (q_C, q_D, q_N) = (4, 16, 180),$$

where S_C contains common natural-parameter coefficients, S_D contains centered component contrasts, and S_N is zero. The clustering difficulty is calibrated by the oracle Bayes error

$$e_B = Pr_{\Theta^\star}\left\{\arg\max_k \Pr(Z = k \mid X; \Theta^\star) \neq Z\right\}.$$

We use $e_B \in \{0.025, 0.05, 0.10\}$, $n \in \{300, 1000\}$, and either

$$\kappa = (45, 45, 45, 45) \quad \text{or} \quad \kappa = (30, 40, 50, 60).$$

The common-to-contrast allocation is calibrated by Monte Carlo bisection while the prescribed concentrations and support sets are held fixed. Independent validation samples of size 200,000 gave achieved errors between 0.02566 and 0.10035; the largest absolute discrepancy from its target was 0.00172.

Each of the 12 cells contains 100 paired repetitions and an independent test sample of size 5,000. The common comparison set consists of spherical k-means, dense vMF mixtures with shared and component-specific concentrations, the Rossi-type entrywise prototype lasso (M-L), E-CGL, and E-ACGL. M-CGL is included as a directional companion in the four prespecified $e_B = 0.05$ cells. E-CGL is the primary estimator; E-ACGL is an adaptive sensitivity extension. M-L, E-CGL, and E-ACGL use 240-point paths. M-CGL uses the union of supports from 60- and 120-point paths. All mixture fits use 10 dense starts, Rcpp helpers, and BIC after target-specific support refitting.

Table 1 records the estimand and role of each method. The common comparison set is evaluated in all 12 primary cells; M-CGL is restricted to the four prespecified $e_B = 0.05$ companion cells.

Clustering and predictive fit are evaluated by ARI, NMI, and test negative log-likelihood (NLL). Posterior-score support recovery is summarized by

$$\mathrm{TPR}_\eta, \quad \mathrm{FPR}_\eta, \quad F_{1,\eta}, \quad \Pr(\widehat{S}_\eta = S_\eta),$$

and centered natural-parameter estimation by

$$\mathrm{MSE}_\eta = \frac{1}{Kd}\sum_{k=1}^{K}\sum_{j=1}^{d}(\widehat{c}_{kj} - c^\star_{kj})^2,$$

after label alignment. Each sparse procedure is also evaluated against its own population support. Consequently, $F_{1,P}$, $F_{1,\mu}$, and $F_{1,\eta}$ are target-specific quantities rather than a common ranking.

Table 1: Methods and roles in the numerical studies.

Method	support target	role
Spherical k-means	none	external clustering baseline
Dense vMF, shared κ	none	concentration-restricted baseline
Dense vMF, free κ_k	none	likelihood baseline
M-L	S_P	published prototype-sparsity reference
M-CGL	S_μ	directional companion
E-CGL	S_η	primary proposed estimator
E-ACGL	S_η	adaptive sensitivity extension

4.2. Posterior-score support recovery

Across 11 of the 12 primary cells, E-CGL selected between 16.02 and 16.60 coordinates and attained $F_{1,\eta} \geq 0.983$. The exception combined $n = 300$, $e_B = 0.10$, and heterogeneous concentrations. In that cell E-CGL selected 24.34 coordinates and attained $F_{1,\eta} = 0.768$; E-ACGL selected 15.23 coordinates and attained $F_{1,\eta} = 0.948$. At $n = 1000$, E-CGL had $F_{1,\eta} \geq 0.986$ in all six concentration–overlap cells. ARI decreased with increasing e_B even where support recovery remained accurate, which reflects the calibrated increase in population overlap. Full method-by-cell means and Monte Carlo standard errors are reported in Supplementary Tables S1–S2.

Table 2 gives the representative heterogeneous $e_B = 0.05$ comparison. M-CGL and E-CGL approach their respective 20- and 16-coordinate targets at $n = 1000$, whereas M-L retains nearly all coordinates. The similarity of ARI for M-CGL and E-CGL does not make their supports interchangeable.

4.3. Directional versus posterior-score estimands

Four diagnostics separate directional and posterior-score heterogeneity. Under common concentration, $S_\mu = S_\eta$. Under pure concentration heterogeneity, the component directions are common, so $S_\mu = \varnothing$ while S_η contains 16 coordinates. The shared-canonical design contains 80 common natural-parameter coordinates and 20 posterior-score coordinates. The crossed design contains four η-only, four μ-only, eight jointly active, and eight null coordinates.

For the empty directional target, "exact own target" is the probability of selecting the empty support. M-CGL selected that support in 24% of the pure- concentration repetitions, whereas E-CGL recovered the nonempty posterior-score support with mean $F_{1,\eta} = 0.990$. In the crossed design, each

Table 2: Representative comparison at $e_B = 0.05$ with heterogeneous concentrations. Support F1 is evaluated against each method's own estimand.

n	method	selected q	target F1	ARI	test NLL
300	spherical k-means	–	–	0.580	–
300	dense shared-κ	–	–	0.649	-245.860
300	dense free-κ_k	–	–	0.709	-246.082
300	M-L	199.70	0.182	0.718	-246.122
300	M-CGL	20.73	0.897	0.812	-247.060
300	E-CGL	16.60	0.983	0.857	-247.206
300	E-ACGL	16.18	0.995	0.858	-247.215
1000	spherical k-means	–	–	0.773	–
1000	dense shared-κ	–	–	0.765	-247.004
1000	dense free-κ_k	–	–	0.835	-247.228
1000	M-L	199.67	0.182	0.836	-247.232
1000	M-CGL	20.04	0.999	0.867	-247.510
1000	E-CGL	16.02	0.999	0.869	-247.517
1000	E-ACGL	16.02	0.999	0.869	-247.517

method had high F1 against its own target and lower F1 against the other target. The shared-canonical design exposed a boundary for the directional companion: its exact support did not occur on the recorded path. These results motivate target-specific, rather than winner-takes-all, comparisons.

4.4. Sample-size recovery and oracle benchmarks

Figure 1 follows E-CGL over $n \in \{300, 600, 1000, 2000\}$. The $e_B = 0.05$ trajectories include equal and heterogeneous concentrations; the $e_B = 0.10$ trajectory uses heterogeneous concentrations. At $e_B = 0.05$, centered-η MSE decreased from 0.221 to 0.028 under equal concentrations and from 0.229 to 0.028 under heterogeneous concentrations. FPR decreased to zero and exact-support recovery reached one at $n = 2000$. Under $e_B = 0.10$ and heterogeneous concentrations, E-CGL moved from $\widehat{q} = 24.34$, $F_{1,\eta} = 0.768$, and exact recovery 0.01 at $n = 300$ to $\widehat{q} = 16.03$, $F_{1,\eta} = 0.999$, and exact recovery 0.97 at $n = 2000$.

The path oracle is the fitted candidate with the largest target-specific F1 within each repetition. The oracle-S_η estimator refits under the known population support. We report

$$\Delta_{\text{sel}} = F_1^{\text{path oracle}} - F_1^{\text{BIC}}, \qquad \Delta_{\text{NLL}} = \text{NLL}(\widehat{\Theta}) - \text{NLL}(\widehat{\Theta}_{S_\eta}^{\text{oracle support}}).$$

Table 3: Estimand-separation diagnostics over 100 repetitions.

design	method	$\widehat{q}$	$F_{1,\mu}$	$F_{1,\eta}$	exact own target	ARI
common κ	M-CGL	16.04	0.999	0.999	0.96	0.867
common κ	E-CGL	16.06	0.998	0.998	0.95	0.867
pure concentration	M-CGL	0.77	–	0.004	0.24	0.674
pure concentration	E-CGL	16.33	0.000	0.990	0.77	0.635
shared canonical	M-CGL	21.16	0.348	0.973	0.00	0.859
shared canonical	E-CGL	20.02	0.333	1.000	0.98	0.870
crossed support	M-CGL	11.67	0.983	0.677	0.64	0.996
crossed support	E-CGL	11.87	0.681	0.980	0.69	0.996

These finite-sample benchmark gaps are not claims about asymptotic support recovery. At $e_B = 0.05$, all E-CGL selector gaps were at most 0.0079 at $n = 300$ and at most 0.0018 for $n \geq 600$. The difficult $e_B = 0.10$ trajectory had $\Delta_{\text{sel}} = 0.046$ and $\Delta_{\text{NLL}} = 0.191$ at $n = 300$; both gaps narrowed with sample size. The full 12-row trajectories and all BIC, path-oracle, and oracle-support-refit results are in Supplementary Tables S3–S4.

4.5. Stress conditions and computation

Five stress cells examine moderately dense support at $n = 300$ and $n = 1000$, strongly dense support, $d > n$, and a weak beta-min condition. In the sparse weak-signal cell, E-CGL retained mean $F_{1,\eta} = 0.998$ and exact recovery 0.93. In the $d = 500, n = 300$ cell, its F1 was 0.804 and exact recovery 0.01. For 80 active coordinates, E-CGL F1 increased from 0.792 at $n = 300$ to 0.890 at $n = 1000$, but exact recovery remained zero. With 160 active coordinates, E-CGL selected 153.96 on average and had $F_{1,\eta} = 0.924$; E-ACGL selected 125.69 and had $F_{1,\eta} = 0.878$. Thus adaptive weighting was useful in some high-dimensional and moderately dense cells but not in the strongly dense negative control. Full stress results and method-phase convergence summaries are reported in Supplementary Tables S5–S6.

In a paired runtime diagnostic for the representative heterogeneous $e_B = 0.05$, $n = 1000$, $d = 200$ cell, median end-to-end wall-clock times (IQR) were 383.02 (343.71–418.68) seconds for M-CGL, 48.31 (46.36–49.85) seconds for E-CGL, and 43.14 (42.78–43.58) seconds for E-ACGL. All 70 method–replicate fits were valid. The sparse-method totals include the paired dense initialization and the method-specific path, refit, and selector; one-time Rcpp

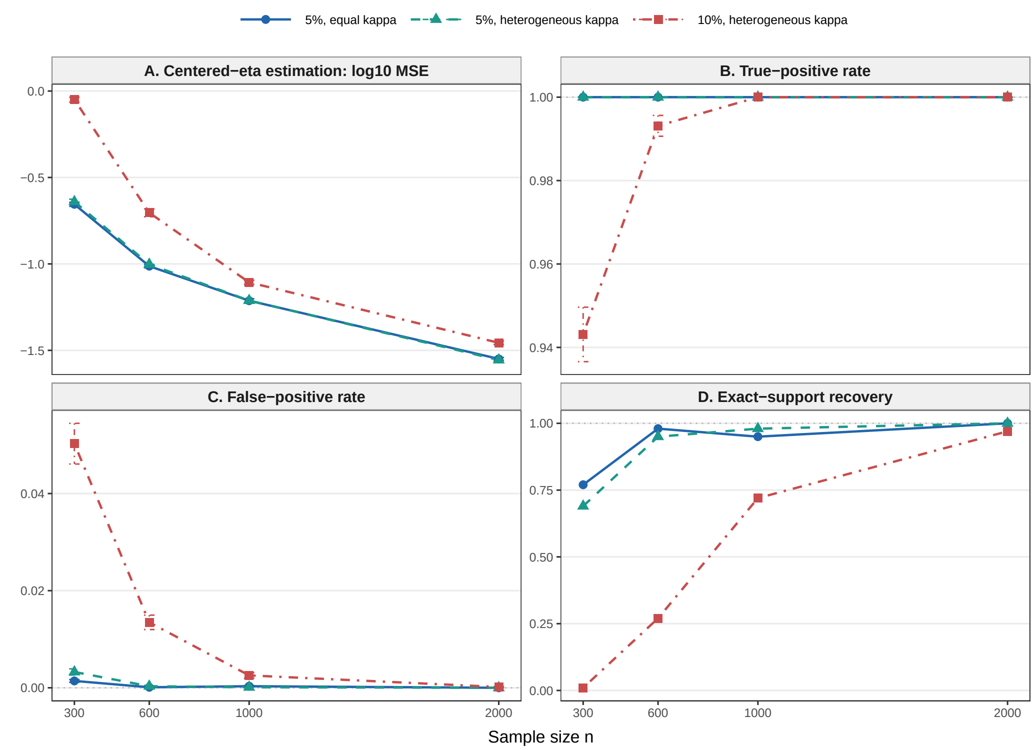

Figure 1: E-CGL sample-size trajectories. Points are Monte Carlo means and error bars are Monte Carlo standard errors over 100 repetitions.

compilation was excluded. Because the methods use different path and optimization rules, these values describe computational burden rather than establishing speed superiority. The full seven-method timing panel, iteration counts, failure status, and computing environment are reported in the matched-runtime table in the Supplement.

The frozen Study A, sample-size, and stress outputs contain no error rows or duplicate method–replicate keys. Two standard-method fits reported nonconvergence. Selected E-series fits passed the recorded majorization, line-search, and path-endpoint gates. Objective, residual, R/Rcpp-equivalence, and warning-handling diagnostics are reported in the Supplement. These checks are numerical diagnostics and do not establish global optimality.

5. Real data analysis

5.1. Data and frozen analysis protocol

Classic3 contains abstracts from the CISI, CRAN, and MED document collections and has been used in directional document clustering (Banerjee et al., 2005; Greene and Cunningham, 2006). We removed two exact duplicates and five prespecified near duplicates, leaving $n = 3{,}883$ documents. The cleaned class counts were 1,454, 1,397, and 1,032, respectively.

Documents were encoded with the vocabulary-aligned SPLADE CoCondenser model at a fixed model revision (Formal et al., 2021). Coordinates occurring in fewer than two documents were removed, the top 2,000 coordinates were ranked by variance over the unlabeled cleaned payload, and each vector was normalized to unit Euclidean norm. The resulting matrix had no zero rows, duplicate transformed rows, duplicate vocabulary entries, or nonfinite values. The model identifier, revision, vocabulary hash, duplicate audit, and payload hash were frozen before model fitting.

The benchmark component count was fixed at $K = 3$. Labels were used only to verify this prespecified resolution, calculate ARI and NMI after fitting, and name fitted components for interpretation. They were not used for coordinate ranking, initialization, parameter estimation, tuning, or support selection. Five prespecified 80/20 splits are retained in the Supplement as stability and held-out diagnostics; they are not pooled with the full-data estimates.

5.2. Methods and evaluation

The comparison comprised spherical k-means, permutation-tuned sparse k-means, dense vMF mixtures with shared and component-specific concentrations, the Rossi-style M-L prototype selector, M-CGL, E-CGL, and the adaptive E-ACGL extension. Likelihood-based fits used 30 dense random starts. Centered paths used 240 candidates, whereas the M-L path allowed 600 updates. The sparse vMF methods were selected by their method-specific BIC after support-constrained refitting. Sparse k-means used a label-blind permutation gap criterion with 25 permutations and 10 weight-bound candidates.

Selected q is target-specific: weighted coordinates for sparse k-means, prototype-union coordinates for M-L, centered directional coordinates for M-CGL, and centered natural-parameter coordinates for E-CGL and E-ACGL. Accordingly, differences in q do not by themselves rank the estimands. All

Table 4: Classic3 full-data results after duplicate removal ($n = 3{,}883$, $d = 2{,}000$, $K = 3$). Component order is aligned to CISI/CRAN/MED only after fitting.

Method	q (q/d)	Cluster sizes	ARI	NMI	ℓ	BIC	$\widehat{\kappa}$
Spherical k-means	2000 (1.000)	1458/1406/1019	0.970	0.944	–	–	–
Sparse k-means	985 (0.492)	553/784/2546	0.137	0.313	–	–	–
Dense vMF shared-kappa	2000 (1.000)	1458/1406/1019	0.970	0.944	18,917,075	-37,784,564	732.2/732.2/732.2
Dense vMF free-kappa	2000 (1.000)	1462/1391/1030	0.976	0.954	18,921,337	-37,793,072	738.5/818.0/615.3
M-L	1942 (0.971)	1462/1405/1016	0.970	0.944	18,914,871	-37,804,965	730.0/730.0/730.0
M-CGL	1462 (0.731)	1456/1392/1035	0.976	0.954	18,919,418	-37,798,125	738.8/816.4/609.9
E-CGL	1421 (0.711)	1456/1392/1035	0.976	0.954	18,919,122	-37,798,211	738.3/815.9/610.4
E-ACGL	2000 (1.000)	1462/1391/1030	0.976	0.954	18,921,337	-37,793,072	738.4/818.0/615.3

Selected q refers to the method-specific target: weighted coordinates for sparse k-means, prototype-union coordinates for M-L, centered directional coordinates for M-CGL, and centered natural-parameter coordinates for E-CGL/E-ACGL. Likelihood and BIC are not defined for the two k-means procedures. $\widehat{\kappa}$ is ordered as CISI/CRAN/MED after post-fit component alignment.

methods were evaluated by ARI and NMI. Observed log-likelihood and the practical nominal-dimension BIC were additionally reported for vMF models.

5.3. Full-data results and interpretation

The dense free-κ_k mixture attained ARI 0.9760 and NMI 0.9538. E-CGL selected 1,421 of 2,000 coordinates, excluding 28.95% of the coordinates from posterior-score contrasts while attaining ARI 0.9761 and NMI 0.9539. M-CGL gave the same partition with 1,462 directional-contrast coordinates. M-L retained 1,942 prototype coordinates, and E-ACGL selected the dense support. Thus the adaptive extension did not add sparsity in this application. The permutation-tuned sparse k-means fit selected 985 coordinates but had ARI 0.1374; its selected weight bound was not at the evaluated grid boundary.

The E-CGL concentrations, aligned to CISI, CRAN, and MED only after fitting, were 738.31, 815.88, and 610.41. Figure 2 ranks tokens by the fitted centered natural-parameter coefficient multiplied by the empirical root-mean-square feature magnitude. Positive CISI contrasts were led by *library*, *information*, and *librarian*; CRAN by *flow*, *mach*, and *pressure*; and MED by *tumor*, *inhibitor*, and *rat*. The corresponding negative contrasts show which vocabularies oppose each fitted component. Coordinates excluded from the contrast support may retain a nonzero common natural-parameter baseline; their exclusion does not imply absence from the fitted density.

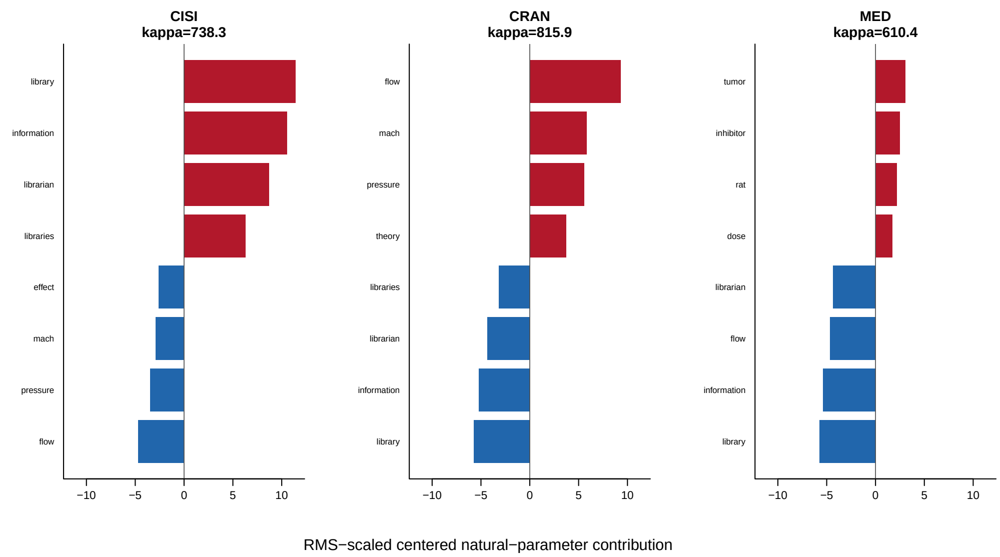

Figure 2: Leading positive (red) and negative (blue) E-CGL centered natural-parameter contributions in Classic3. Contributions are scaled by the empirical root-mean-square feature magnitude. Labels were used only after fitting to name components.

5.4. Supplementary stability and scope checks

The Supplement reports the five prespecified Classic3 splits, support Jaccard indices and token-selection frequencies, and the BBCSport contrast analysis. It also retains CSTR as a Rossi-style implementation bridge. In the Classic3 splits, E-CGL had mean conditional support Jaccard 0.933. BBCSport illustrates a setting in which coordinate reduction incurred held-out density loss, while CSTR is used only to check comparability with the published prototype-sparse analysis. Real data provide no true coordinate support, so support TPR, FPR, precision, and F1 are not reported.

6. Discussion

6.1. Posterior-score contrast support as the estimand

This study defines posterior-score contrast support as the coordinates entering the feature-dependent linear term of pairwise posterior log-score contrasts. E-CGL regularizes centered natural-parameter contrasts and therefore targets posterior-score coordinates. The M-L, M-GL, and M-AGL methods target prototypes. The M-CGL companion targets centered directional support. Centering leaves the common natural-parameter baseline unpenalized, so a coordinate shared across components is not selected solely because it remains nonzero in every prototype. The coordinate-group penalty is justified by alignment between the regularization unit and this estimand, rather than by uniform empirical superiority over entry-wise regularization or prototype-oriented penalties.

6.2. Empirical scope and adaptive extension

Across the evaluated $n = 1000$ Study B cells, E-CGL selected 16.01–16.16 coordinates on average and attained $F_{1,\eta}$ between 0.995 and 1.000 for a true support size of 16 (Supplementary Tables S1–S2). Clustering ARI decreased with increasing oracle Bayes error, whereas support recovery changed little in these cells; clustering and support difficulty therefore need not vary in parallel in the evaluated settings. The shared-background experiment illustrated the practical difference between prototype and posterior-score contrast support. The dense-support experiment, in contrast, documented substantial finite-sample under-selection when decision information was distributed across many weak coordinates, although recovery improved at $n = 1000$ in that design.

E-ACGL improved false-positive control in some difficult small-sample cells but was otherwise similar to E-CGL (Supplementary Tables S1–S2). We therefore retain E-CGL as the primary estimator and treat E-ACGL as an adaptive extension whose performance depends on the initial contrast estimates. In the final-engine matched ablation, the three centered penalties recovered the same support in the well-separated cell. In the difficult small-sample cell, E-CGL reduced noise selection relative to E-CL, and E-ACGL reduced it further. This result supports coordinate grouping in that regime but does not establish uniform superiority; the primary motivation remains the coordinate-level definition of the support target.

A supplementary target-separation diagnostic evaluated M-CGL and E-CGL against centered directional and centered natural-parameter support, respectively. Both methods recovered their targets in the common- concentration and common-natural-parameter designs. In the pure-concentration design, E-CGL recovered the nonempty canonical support with mean target F1 0.989, whereas M-CGL selected the correct empty directional support in 9 of 20 repetitions. The latter result documents path and numerical sensitivity at the empty-support boundary and is one reason M-CGL is retained as a directional diagnostic rather than a co-primary estimator.

The locked real-data analyses provided complementary evidence through held-out clustering and support stability rather than recovery of an unknown true support. In Classic3, E-CGL excluded 32.9% of the coordinates, with mean ARI and NLL costs of 0.0036 and 0.6705 per document relative to dense free-κ_k vMF (see the five-split results in the Supplement). Its conditional support Jaccard was 0.933, and the signed token contrasts were coherent across the five splits. In BBCSport, every sparse method had higher held-out NLL than its matched dense model in all five splits. CSTR is retained only as an implementation bridge for the Rossi-style method because its earlier E-series diagnostic used a different selector. These results delimit the intended use of E-CGL rather than establish a uniform ordering of support targets.

6.3. Model selection and computation

The support-recovery claims condition on a fixed number of components. The separate two-stage K diagnostic in Section 3.3.4 is a practical procedure and does not establish consistency of component-number selection. In Classic3, held-out density favored finer component resolutions, whereas partition stability was maximized at the three supplied broad topics. The number of fitted density components and the resolution of an external label system should therefore be reported separately.

Support selection uses the unpenalized observed likelihood after exact centered-support refitting (Section 3.3.3). The corresponding BIC degrees of freedom are a practical model-selection approximation, not an exact effective-degrees-of- freedom result for a nonregular penalized mixture. Sensitivity analyses showed limited variation among the stored exact-refit candidates under the examined df and EBIC rules, but do not provide a general selection-consistency result.

The fitting algorithm controls accepted generalized-M updates through smooth majorization and backtracking, with a separate step-halving safeguard during exact refitting. These results concern objective control for accepted steps; they do not imply convergence of the full parameter sequence, stationarity, or global optimality. Optional Rcpp helpers reproduced the R-only outputs to numerical tolerance and reduced elapsed time in diagnostic benchmarks. This is an implementation result rather than a statistical performance claim.

6.4. Limitations and future directions

E-CGL is intended for settings in which posterior-score contrast support is sparse or compressible. Weak separation, many weak active coordinates, or small samples can produce under-selected or empty supports, and mixture fitting remains sensitive to initialization and local optima. The main simulations fix the true K to isolate support recovery, while the separate K experiments provide diagnostics rather than an inferential guarantee. Real-data support cannot be evaluated by TPR or F1 because its true coordinate support is unknown; the evidence instead rests on stability, held-out performance, and substantive interpretation. The Classic3 and BBCSport conclusions are also conditional on the fixed pretrained SPLADE revision and training-ranked vocabulary filters. The selected support is coordinate-system dependent: token-level interpretation is meaningful because the representation is aligned with the vocabulary. It does not transfer unchanged to an arbitrarily rotated dense embedding space.

Further work is needed on effective degrees of freedom for centered penalized mixtures, joint uncertainty in K and support, and selection rules that remain stable under diffuse weak signals. Extensions that combine sparse decision support with explicit dense-background components may also reduce the finite-sample cost observed in dense-support settings.

Centered natural-parameter group regularization defines an estimator of sparse posterior-score contrast support in vMF mixtures. Its target differs from a sparse component prototype: it retains coordinates that change posterior component comparisons while leaving common natural-parameter baselines unpenalized. The simulation and Classic3 analyses support this target-specific interpretation, and the dense-support and supplementary real-data diagnostics identify settings in which sparsification can be costly. E-CGL is therefore positioned as a posterior-score-support estimator for sparse or compressible contrast structures, not as a uniformly superior replacement for dense or prototype-oriented mixture methods.

Appendix A. Proofs of propositions

Proof of Proposition 1. Taking the average of $v = \bar{v}\mathbf{1}_K + c$ and using $\mathbf{1}_K^\top c = 0$ determines $\bar{v}$ and hence c uniquely. The three zero-contrast statements are immediate. Finally, $\sum_{k<\ell}(v_k - v_\ell)^2 = K\sum_k (v_k - \bar{v})^2$, which gives the stated identity. $\square$

Proof of Proposition 2. Because $P\mathbf{1}_K = \mathbf{1}_K$, P commutes with J_K and H_K. Permutation matrices are orthogonal and therefore preserve the Euclidean norm. Applying Proposition 3 gives equivariance of the proximal map. □

Proof of Proposition 3. The spaces spanned by $\mathbf{1}_K$ and orthogonal to $\mathbf{1}_K$ are orthogonal. The penalty is zero on the first space and equals the Euclidean norm on the second. The minimization therefore preserves $J_K v$ and applies the standard group soft-thresholding map to $H_K v$. □

Proof of Proposition 4. The vMF log-partition function has Hessian equal to the covariance matrix of a unit-norm random vector and is therefore positive semidefinite with operator norm at most one. The Hessian of the kth block of f_t consequently has operator norm at most N_k. The descent result follows by combining the smooth majorization inequality with optimality of the proximal update. □

Proof of Proposition 5. The standard EM lower-bound identity expresses the observed log-likelihood difference as the auxiliary-function difference plus a nonnegative Kullback–Leibler divergence term. The penalty depends only on the parameters and can be subtracted from both sides, yielding the penalized result. □

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